

## SAFETY DATA SHEET

Creation Date 19-Sep-2014

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Revision Number 5

### 1. Identification

**Product Name** Calcium chloride hexahydrate

**Cat No. :** AC389260000; AC389260010; AC389260025; AC389262500

**CAS No** 7774-34-7  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Serious Eye Damage/Eye Irritation

Category 2

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Causes serious eye irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Wear eye/face protection

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                     | CAS No     | Weight % |
|-------------------------------|------------|----------|
| Calcium chloride, hexahydrate | 7774-34-7  | <=100    |
| Calcium chloride              | 10043-52-4 | 0        |

### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Substance is nonflammable; use agent most appropriate to extinguish surrounding fire. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                              |
| <b>Flash Point</b>                    | No information available                                                              |
| <b>Method -</b>                       | No information available                                                              |
| <b>Autoignition Temperature</b>       | No information available                                                              |
| <b>Explosion Limits</b>               |                                                                                       |
| <b>Upper</b>                          | No data available                                                                     |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes.

**Hazardous Combustion Products**

Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions** Should not be released into the environment.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing.

**Storage.** Keep container tightly closed in a dry and well-ventilated place. To maintain product quality: Keep refrigerated. Store under an inert atmosphere. Protect from moisture. Incompatible Materials. Strong oxidizing agents. Strong acids. Metals.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical State</b> | Solid                    |
| <b>Appearance</b>     | White                    |
| <b>Odor</b>           | No information available |

|                                        |                                         |
|----------------------------------------|-----------------------------------------|
| Odor Threshold                         | No information available                |
| pH                                     | No information available                |
| Melting Point/Range                    | 30 °C / 86 °F                           |
| Boiling Point/Range                    | No information available                |
| Flash Point                            | No information available                |
| Evaporation Rate                       | Not applicable                          |
| Flammability (solid,gas)               | No information available                |
| Flammability or explosive limits       |                                         |
| Upper                                  | No data available                       |
| Lower                                  | No data available                       |
| Vapor Pressure                         | No information available                |
| Vapor Density                          | Not applicable                          |
| Specific Gravity                       | No information available                |
| Solubility                             | 5360 g/L (20°C)                         |
| Partition coefficient; n-octanol/water | No data available                       |
| Autoignition Temperature               | No information available                |
| Decomposition Temperature              | No information available                |
| Viscosity                              | Not applicable                          |
| Molecular Formula                      | Ca Cl <sub>2</sub> · 6 H <sub>2</sub> O |
| Molecular Weight                       | 219.08                                  |

## 10. Stability and reactivity

|                                  |                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                |
| Stability                        | Hygroscopic.                                                                              |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Excess heat. Exposure to moist air or water. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Metals                                             |
| Hazardous Decomposition Products | Hydrogen chloride gas                                                                     |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                  |
| Hazardous Reactions              | None under normal processing.                                                             |

## 11. Toxicological information

### Acute Toxicity

|                       |                                                               |
|-----------------------|---------------------------------------------------------------|
| Product Information   | The toxicological properties have not been fully investigated |
| Component Information |                                                               |

| Component        | LD50 Oral          | LD50 Dermal                  | LC50 Inhalation |
|------------------|--------------------|------------------------------|-----------------|
| Calcium chloride | 2301 mg/kg ( Rat ) | LD50 > 5000 mg/kg ( Rabbit ) | Not listed      |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes                                                                       |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                     | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------|------------|------------|------------|------------|------------|------------|
| Calcium chloride, hexahydrate | 7774-34-7  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Calcium chloride              | 10043-52-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                          |
|-------------------|--------------------------|
| Mutagenic Effects | No information available |
|-------------------|--------------------------|

|                                                   |                           |
|---------------------------------------------------|---------------------------|
| <b>Reproductive Effects</b>                       | No information available. |
| <b>Developmental Effects</b>                      | No information available. |
| <b>Teratogenicity</b>                             | No information available. |
| <b>STOT - single exposure</b>                     | None known                |
| <b>STOT - repeated exposure</b>                   | None known                |
| <b>Aspiration hazard</b>                          | No information available  |
| <b>Symptoms / effects, both acute and delayed</b> | No information available  |
| <b>Endocrine Disruptor Information</b>            | No information available  |
| <b>Other Adverse Effects</b>                      | No information available. |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component        | Freshwater Algae | Freshwater Fish                           | Microtox   | Water Flea        |
|------------------|------------------|-------------------------------------------|------------|-------------------|
| Calcium chloride | Not listed       | Lepomis macrochirus: LC50: 10650 mg/L/96h | Not listed | EC50: 52 mg/L/48h |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

### United States of America Inventory

| Component                     | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Calcium chloride, hexahydrate | 7774-34-7  | -    | -                                             | -                           |
| Calcium chloride              | 10043-52-4 | X    | ACTIVE                                        | -                           |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                     | CAS No     | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-------------------------------|------------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Calcium chloride, hexahydrate | 7774-34-7  | -   | -   | -         | X     | -    |      | X    | X     | -        |
| Calcium chloride              | 10043-52-4 | X   | -   | 233-140-8 | X     | X    | X    | X    | X     | KE-04496 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations**

|                                                             |                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------|
| <b>SARA 313</b>                                             | Not applicable                                              |
| <b>SARA 311/312 Hazard Categories</b>                       | See section 2 for more information                          |
| <b>CWA (Clean Water Act)</b>                                | Not applicable                                              |
| <b>Clean Air Act</b>                                        | Not applicable                                              |
| <b>OSHA - Occupational Safety and Health Administration</b> | Not applicable                                              |
| <b>CERCLA</b>                                               | Not applicable                                              |
| <b>California Proposition 65</b>                            | This product does not contain any Proposition 65 chemicals. |
| <b>U.S. State Right-to-Know Regulations</b>                 | Not applicable                                              |

**U.S. Department of Transportation**

|                             |   |
|-----------------------------|---|
| Reportable Quantity (RQ):   | N |
| DOT Marine Pollutant        | N |
| DOT Severe Marine Pollutant | N |

|                                             |                                                  |
|---------------------------------------------|--------------------------------------------------|
| <b>U.S. Department of Homeland Security</b> | This product does not contain any DHS chemicals. |
|---------------------------------------------|--------------------------------------------------|

**Other International Regulations**

|                       |                          |
|-----------------------|--------------------------|
| <b>Mexico - Grade</b> | No information available |
|-----------------------|--------------------------|

**Authorisation/Restrictions according to EU REACH**

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Calcium chloride | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                     | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Calcium chloride, hexahydrate | 7774-34-7  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Calcium chloride              | 10043-52-4 | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                     | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Calcium chloride, hexahydrate | 7774-34-7  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Calcium chloride              | 10043-52-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Creation Date** 19-Sep-2014  
**Revision Date** 25-Dec-2021  
**Print Date** 25-Dec-2021  
**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**